ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 style (design-related), built around one case study: an automated pneumatic drilling station. Section A: objective questions (30) · Section B: the cylinder sequence, displacement-step diagram and cylinder sizing, electro-pneumatic and PLC control with ladder logic, timers and sensor wiring, sensor selection, safety, maintenance and fault finding (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In the sequence A+ B+ B− A−, "A+" means that …

A cylinder A retracts. B cylinder A extends.
C valve A switches off. D sensor A fails.

1.1.2 The diagram that shows the movement of each cylinder against the steps of a sequence is the …

A circuit diagram. B ladder diagram.
C displacement-step diagram. D block diagram.

1.1.3 The proximity sensor that detects only metal objects is the …

A capacitive sensor. B ultrasonic sensor. C photoelectric sensor. D inductive sensor.

1.1.4 The sensor that detects a magnet built into a cylinder's piston is a …

A reed switch. B LDR. C thermocouple. D strain gauge.

1.1.5 The PLC timer that switches its output on only after a set delay is the …

A TOF. B CTU. C TON. D RST.

1.1.6 Two-hand control on a machine makes sure that …

A the operator's hands are clear of the danger area. B the machine runs faster.
C two machines start together. D the PLC saves energy.

1.1.7 The output of a PNP proximity sensor is …

A sinking. B sourcing. C analogue only. D a relay.

1.1.8 Lock-out/tag-out is used to …

A speed up maintenance.
B store tools safely.
C prevent a machine from being started during maintenance.
D test sensors.

1.1.9 1 kPa is equal to …

A 1 000 Pa. B 100 Pa. C 10 Pa. D 1 Pa.

1.1.10 A sensor whose output changes smoothly over a range of values is …

A digital. B analogue. C binary. D on/off.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 In the sequence A+ B+ B− A−, cylinder B extends after cylinder A has extended.

1.2.2 Limit switches tell the controller where a cylinder is.

1.2.3 A seal-in (latching) contact keeps a motor running after the start button is released.

1.2.4 The stop button in a ladder rung is normally drawn as a normally open contact.

1.2.5 A capacitive sensor can detect non-metal objects such as plastic and wood.

1.2.6 A through-beam photoelectric sensor has its emitter and receiver in one housing.

1.2.7 An inductive sensor detects stainless steel from further away than mild steel.

1.2.8 A leaking piston seal can make a cylinder move slowly.

1.2.9 Emergency-stop buttons should be wired as normally closed contacts.

1.2.10 Guards may be removed while a machine runs so that the operator can see better.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	List of every PLC input and output with its address	A	hysteresis
1.3.2	Keeps an output waiting for a set time before it switches on	B	limit switch
1.3.3	Detects metal parts without touching them	C	I/O allocation table
1.3.4	Switch that a cylinder operates at the end of its stroke	D	NPN (sinking) output
1.3.5	Light sensor that works with a reflector on the opposite side	E	retro-reflective sensor
1.3.6	Difference between a sensor's switch-on and switch-off points	F	monostable (spring-return) valve
1.3.7	Valve that returns to its normal position by itself	G	inductive sensor
1.3.8	Locking the energy source and fitting a warning label	H	lock-out/tag-out

1.3.9	Planned list of inspections and services	I	on-delay timer
1.3.10	Sensor output that switches 0 V to the load	J	maintenance checklist

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. A small factory wants an automated drilling station. A pneumatic cylinder A clamps the workpiece, then a pneumatic cylinder B lowers the drill into the part. When the hole is drilled, B retracts and then A unclamps. Each cylinder has limit switches at both ends (a0, a1, b0, b1) and both cylinders are controlled by 5/2 single-solenoid, spring-return valves and a PLC.

QUESTION 2: SEQUENCE AND CYLINDER DESIGN

2.1 Write the sequence of the station in short (+/−) notation and explain what it means. (2)

2.2 Draw the displacement-step diagram for the sequence. (6)

2.3 State what EACH of the limit switches a0, a1, b0 and b1 tells the PLC. (4)

2.4 Give THREE safety requirements for the design of the station. (3)

2.5 The clamp must press with a force of 800 N at a working pressure of 600 kPa. Calculate the minimum bore diameter of cylinder A, and choose a suitable standard bore size from 32 mm, 40 mm, 50 mm and 63 mm. (3)

[18]

QUESTION 3: ELECTRO-PNEUMATIC AND PLC CONTROL

3.1 Explain how a 5/2 single-solenoid, spring-return valve controls a double-acting cylinder. (2)

3.2 Draw up an I/O allocation table for the station, with a start button, the four limit switches and the two valve solenoids. (4)

3.3 The chip conveyor next to the station has a start button (NO), a stop button (NC), a guard switch that opens when the guard is lifted, and a motor overload contact (NC). Draw the ladder rung that starts the conveyor motor and keeps it running (seal-in) until the stop button is pressed, the guard is opened or the motor overloads. (5)

3.4 Explain how a TON timer in the PLC can keep the drill at the bottom of the hole for 3 seconds before B retracts. (3)

3.5 Explain the difference between NPN (sinking) and PNP (sourcing) sensors when they are connected to a PLC input card. (4)

[18]

QUESTION 4: SENSOR SELECTION

4.1 Choose the most suitable sensor for EACH task below and give a reason: (a) detecting that a steel workpiece is in the clamp; (b) detecting plastic workpieces on the conveyor; (c) detecting any part that breaks a beam across a 2 m wide conveyor; (d) detecting the end positions of the piston inside cylinder B. (8)

4.2 An inductive sensor has a rated sensing distance of 8 mm for mild steel. The correction factor for stainless steel is 0,7. Calculate the sensing distance for stainless steel. (2)

4.3 Explain the difference between an analogue sensor and a digital (on/off) sensor, and give ONE example of each from this station. (4)

4.4 Explain what hysteresis is in a proximity sensor and why it is useful. (2)

[16]

QUESTION 5: SAFETY, MAINTENANCE AND FAULT FINDING

5.1 Name FOUR safety features that should be fitted to the drilling station. (4)

5.2 Describe, in order, FOUR steps of a lock-out/tag-out procedure before a technician replaces the drill bit. (4)

5.3 Cylinder B now extends very slowly. Give FOUR possible causes. (4)

5.4 Draw up a weekly preventive-maintenance checklist with FOUR items for the pneumatic system of the station. (4)

5.5 State ONE duty that the Occupational Health and Safety Act places on the employer of the machine operator. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER